Retail Orders Analysis

End-to-end data analytics project with Python (pandas) and SQL

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9,994 orders, 2022-2023 | github.com/gauravkal006

2,215,859	205,169	9.26%	+2.25%
Total sales	Total profit	Profit margin	Sales growth 2023 vs 2022

Project overview

Turn a raw retail orders file into answers about products, months, regions and profit.

Data: Kaggle 'Retail Orders' - 9,994 order lines, United States, Jan 2022 to Dec 2023.

Flow from Ankit Bansal's tutorial

1 Clean

Fix missing values, rename columns, derive discount, sale price and profit in pandas

2 Load

Load the clean table into a SQLite database

3 Analyse

Answer five business questions in SQL

Added by me

- Six more SQL queries using window functions and CTEs
- Charts, an Excel dashboard, this deck and a PDF report
- A one-command pipeline (src/pipeline.py) and checks that SQL and pandas agree

Step 1: Cleaning the data with pandas

- **Missing values:** 'Not Available' and 'unknown' read as NaN (6 ship modes).
- **Column names:** converted to snake_case for pandas and SQL.
- **New columns:** discount, sale_price and profit.
- **Dates:** order_date converted from text to a date.
- **Dropped:** list_price and cost_price (already captured).
- **Result:** 9,994 rows, no duplicate orders, no rows lost.

In [5]:
```
df["discount"] = df["list_price"] * df["discount_percent"] * 0.01
df["sale_price"] = df["list_price"] - df["discount"]
df["profit"] = df["sale_price"] - df["cost_price"]
df[["list_price", "discount_percent", "discount", "sale_price", "cost_price", "profit"]].head()
```

Out[5]:

	list_price	discount_percent	discount	sale_price	cost_price	profit
0	260	2	5.2	254.8	240	14.8
1	730	3	21.9	708.1	600	108.1
2	10	5	0.5	9.5	10	-0.5
3	960	2	19.2	940.8	780	160.8
4	20	5	1.0	19.0	20	-1.0

Notebook: deriving discount, sale_price and profit

Checks after cleaning

In [8]:
```
checks = pd.DataFrame({
    "check": ["rows", "columns", "duplicate order_id", "missing values (all columns)",
              "missing ship_mode", "negative profit lines", "zero-value lines (sale_price = 0)"],
    "before (raw)": [len(raw_default), raw_default.shape[1], raw_default["Order Id"].duplicated().sum(),
                     raw_default.isna().sum().sum(), raw_default["Ship Mode"].isna().sum(), "n/a", "n/a"],
    "after (clean)": [len(df), df.shape[1], df["order_id"].duplicated().sum(),
                      df.isna().sum().sum(), df["ship_mode"].isna().sum(),
                      (df["profit"] < 0).sum(), (df["sale_price"] == 0).sum()],
})
checks
```

Out[8]:

	check	before (raw)	after (clean)
0	rows	9994	9994
1	columns	16	17
2	duplicate order_id	0	0
3	missing values (all columns)	1	6
4	missing ship_mode	1	6
5	negative profit lines	n/a	3997
6	zero-value lines (sale_price = 0)	n/a	507

Notebook: checks before and after cleaning

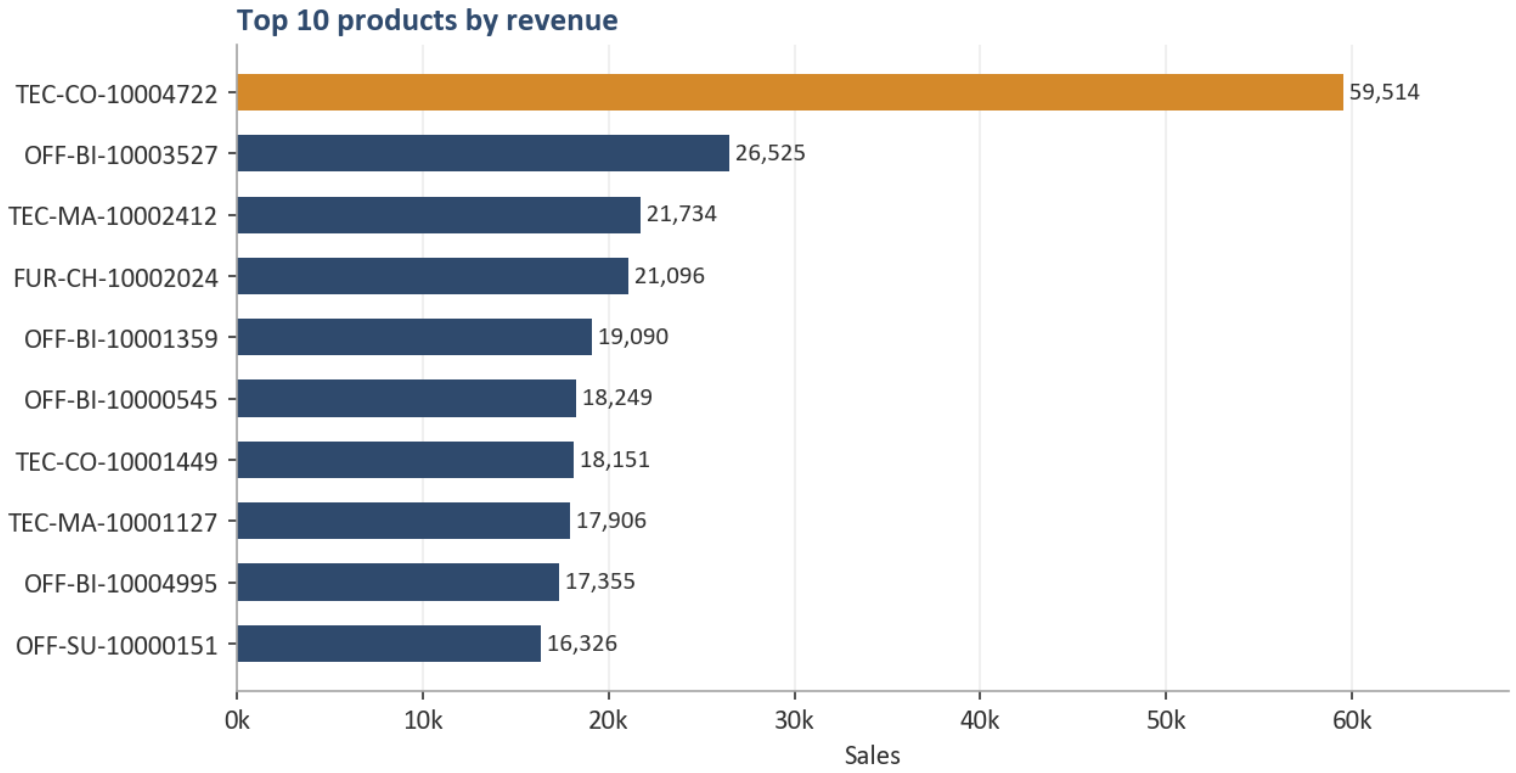
- No rows lost, every order_id unique.
- 507 lines have a price and cost of 0. They add nothing, so they are kept.
- **3,997 lines (40%) lose money.**
  - 3,955 have list price = cost, so any discount makes a loss
  - Total loss is 3,151 (1.5% of profit)

Headline numbers

2,215,859 Total sales	**205,169** Total profit	**9.26%** Profit margin	**9,994** Orders
221.72 Average order value	**80,331** Total discount given	**+2.25%** Sales growth 2023 vs 2022	**+1.90%** Profit growth 2023 vs 2022

- Sales rose from 1,095,590 in 2022 to 1,120,268 in 2023.
- Profit grew slightly slower than sales, so the margin dipped from 9.28% to 9.24%.
- About 1 in every 11 dollars of sales is profit.

Q1. Top 10 products by revenue



- **TEC-CO-10004722** (a copier) earns 59,514, more than twice the second product.
- It is 2.7% of all sales on its own.
- Four of the top ten are binders; two are machines.

Q2 and Q4. Leaders by region and best month by category

Q2: No. 1 product in each region

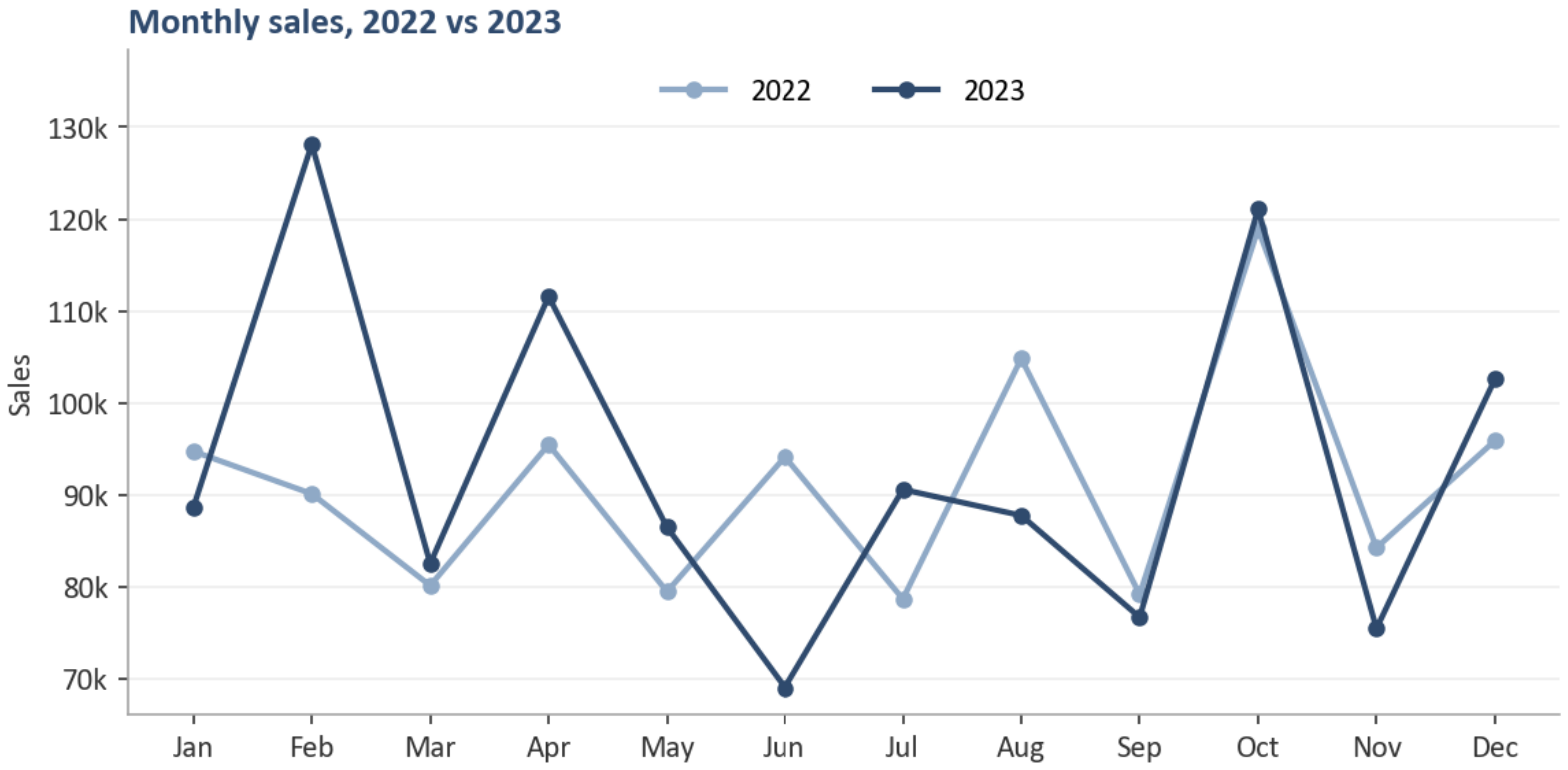
Region	No. 1 product	Sales
Central	TEC-CO-10004722	16,975
East	TEC-CO-10004722	29,099
South	TEC-MA-10002412	21,734
West	TEC-CO-10004722	13,440

Q4: Highest-sales month by category

Category	Best month	Sales
Furniture	2022-10	42,889
Office Supplies	2023-02	44,118
Technology	2023-10	53,000

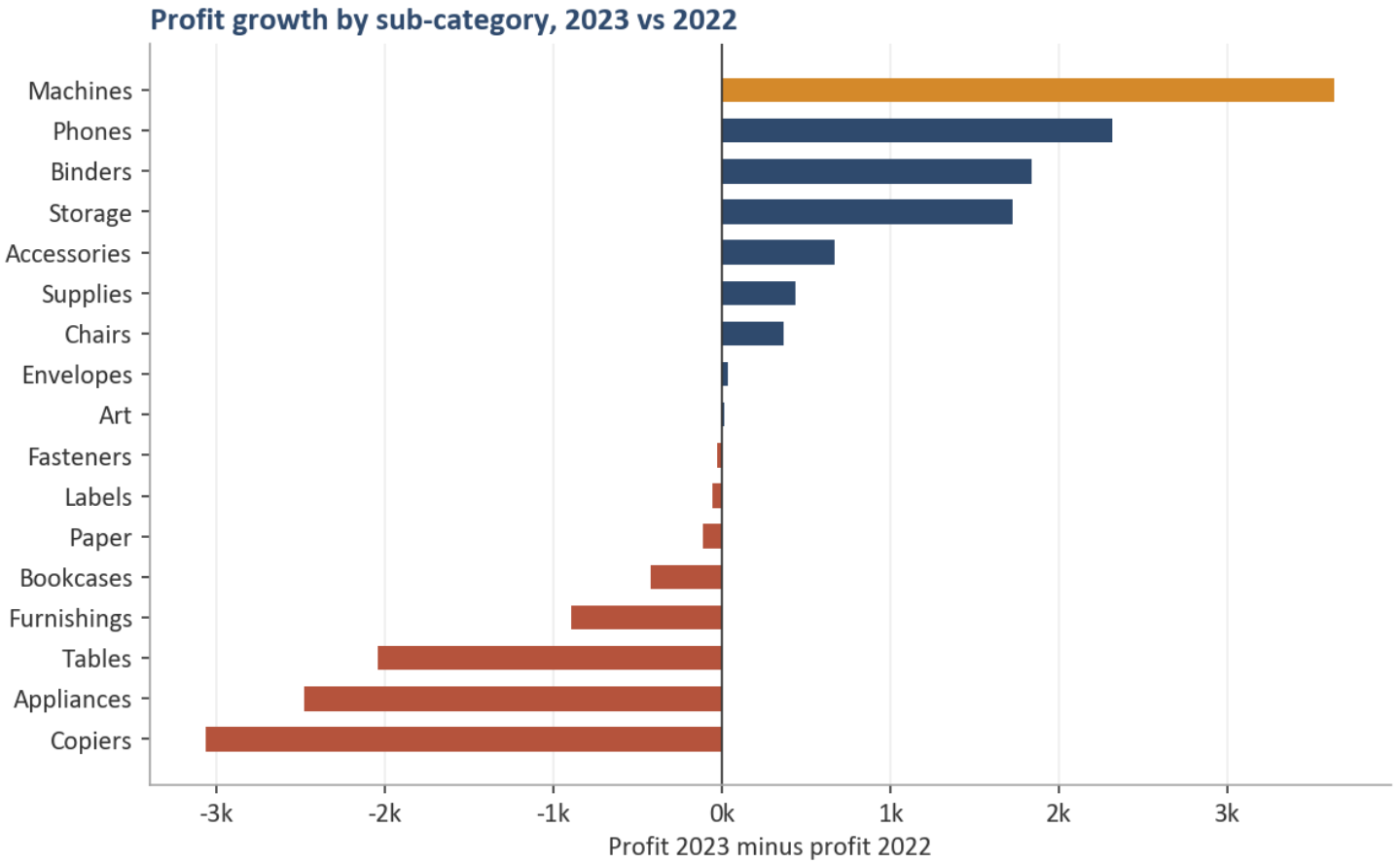
- The same copier leads three of the four regions. The South is led by a machine.
- October appears twice: it is a strong month for Furniture and Technology.
- Full top-5 lists for every region are in sql/analysis.sql (Q2).

Q3. Month-over-month sales, 2022 vs 2023



- 2023 beat 2022 in 7 of 12 months.
- **Biggest gains:** February (+42%) and April (+17%).
- **Biggest falls:** June (-27%) and August (-16%).
- October was a top-two month in both years.

Q5. Which sub-category grew profit the most?



- **Machines:** profit up from 7,243 to 10,878 (+50%).
- Phones, Binders and Storage also grew strongly.
- Copiers, Appliances and Tables lost the most, about 7.6k together.
- 9 sub-categories grew, 8 declined.

Step 2: SQL in SQLite - techniques used

- **CTEs (WITH)** to break queries into steps
- **ROW_NUMBER, RANK, DENSE_RANK** for top-N per group
- **SUM() OVER (PARTITION BY ... ORDER BY ...)** for running totals
- **SUM(SUM(x)) OVER ()** for share of total
- **CASE WHEN** to pivot 2022 and 2023 into columns
- **strftime()** for year and month in SQLite

```
In [21]: run("q9_region_profitability")
```

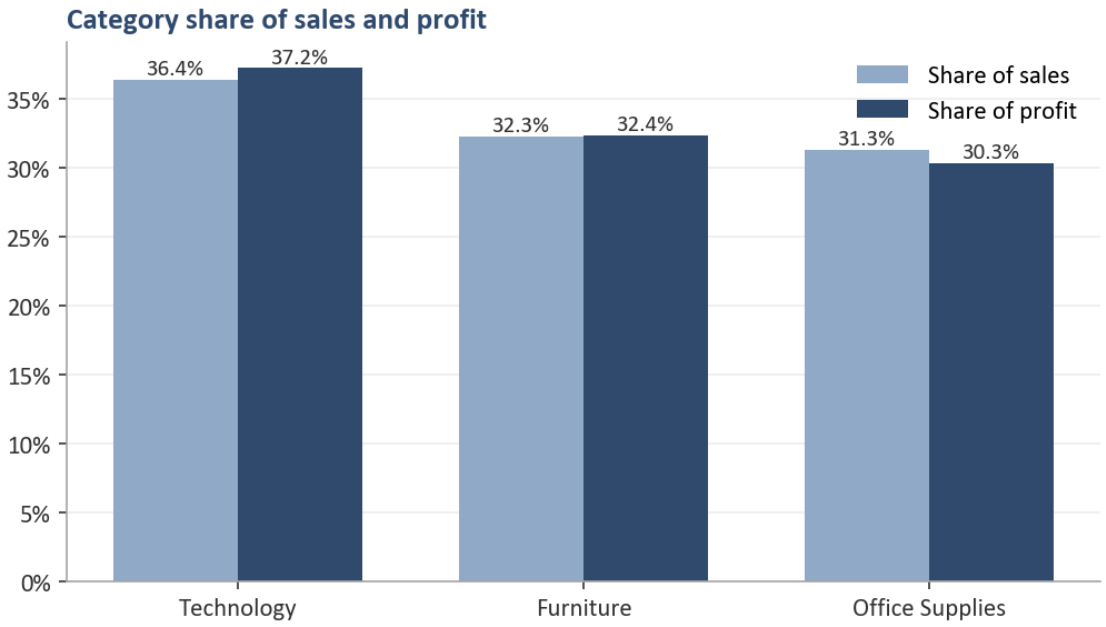
```
-- Q9. Regional sales, profit and margin, with year-over-year sales growth.
```

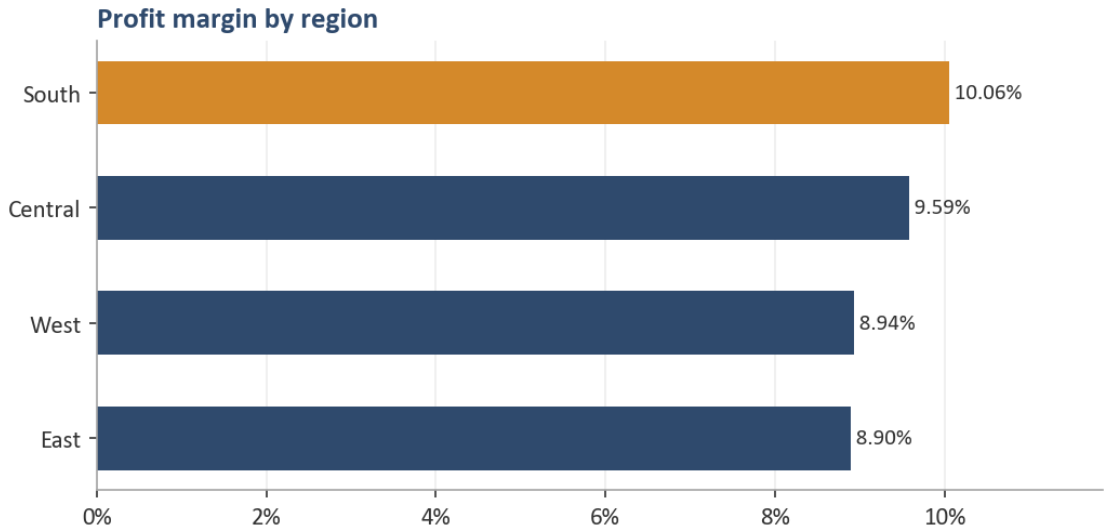
Out[21]:

	region	sales	profit	margin_pct	sales_growth_pct	profit_rank
0	West	699858.6	62558.6	8.94	-11.20	1
1	East	654320.0	58220.0	8.90	11.06	2
2	Central	483668.6	46378.6	9.59	13.09	3
3	South	378011.5	38011.5	10.06	1.17	4

Notebook: Q9 regional profitability query result

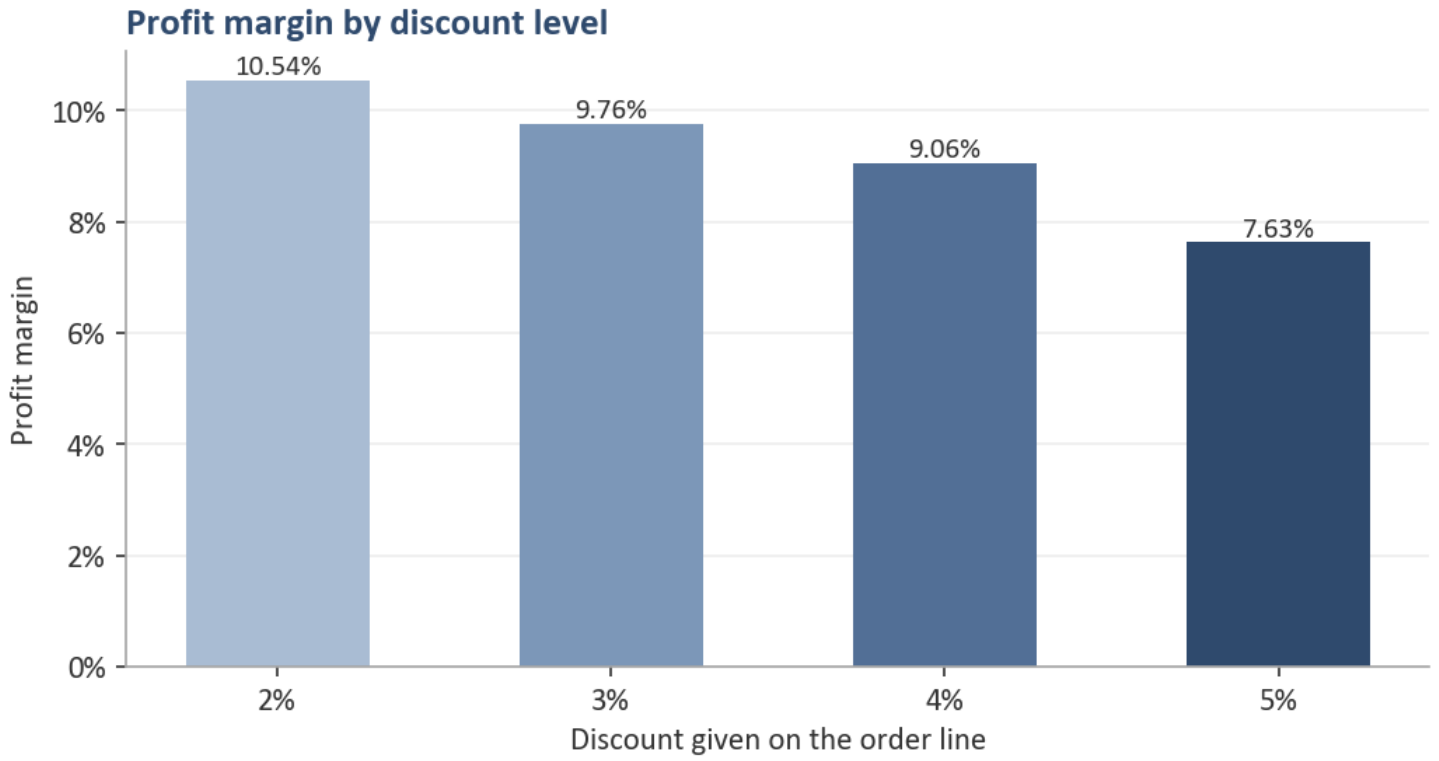
Categories and regions





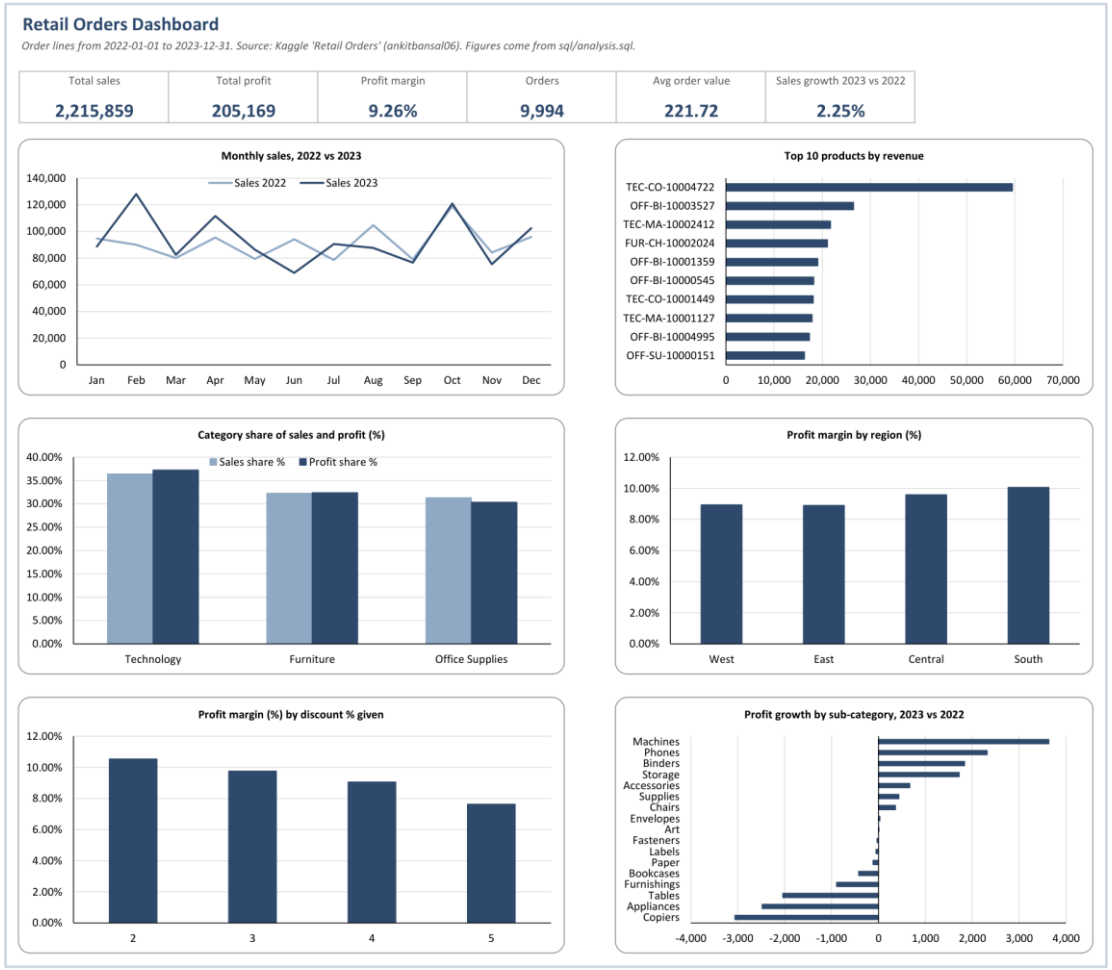
- Technology is the largest category (36% of sales, 37% of profit). Margins are similar across categories.
- South has the best margin (10.06%) but the lowest sales.
- West sells the most, but its sales fell 11.2% in 2023 while the other regions grew.

Discounts reduce margin



- Margin falls from 10.54% at a 2% discount to 7.63% at 5%.
- Each discount level has about 2,500 lines, so this is not a few large orders.
- The 4% and 5% levels gave 51,404 of discount.
- Items with no markup still get discounts and make a loss.

Excel dashboard



excel/Retail_Orders_Dashboard.xlsx - KPI cells, native Excel charts, and data sheets with conditional formatting

Recommendations

1

Stop discounting items sold at cost

This removes almost all loss-making lines.

2

Review the 4-5% discount levels

They cost about 51k and give the lowest margins.

3

Look into the West

The biggest region, but the only one that shrank in 2023.

4

Protect the top copier

One product brings 2.7% of all sales.

5

Check Copiers, Appliances and Tables

Find out why their profit fell: price, cost or mix.

6

Plan for busy months

October is strong every year; February and April were strong in 2023.

Tools and acknowledgement

- **Python:** pandas (cleaning), matplotlib (charts), openpyxl (Excel), python-pptx (slides), Playwright (PDF)
- **SQL:** SQLite - 11 queries in sql/analysis.sql
- **Notebook:** notebooks/retail_orders_analysis.ipynb, executed with outputs
- **Reproducible:** python src/pipeline.py rebuilds the clean data, SQL results, KPIs and charts

Acknowledgement

- The flow and the five core questions follow Ankit Bansal's YouTube tutorial 'End to End Data Analytics Project (Python + SQL)' and his Kaggle dataset. The code was written fresh, and the extended analysis, dashboard, report and slides were added.